

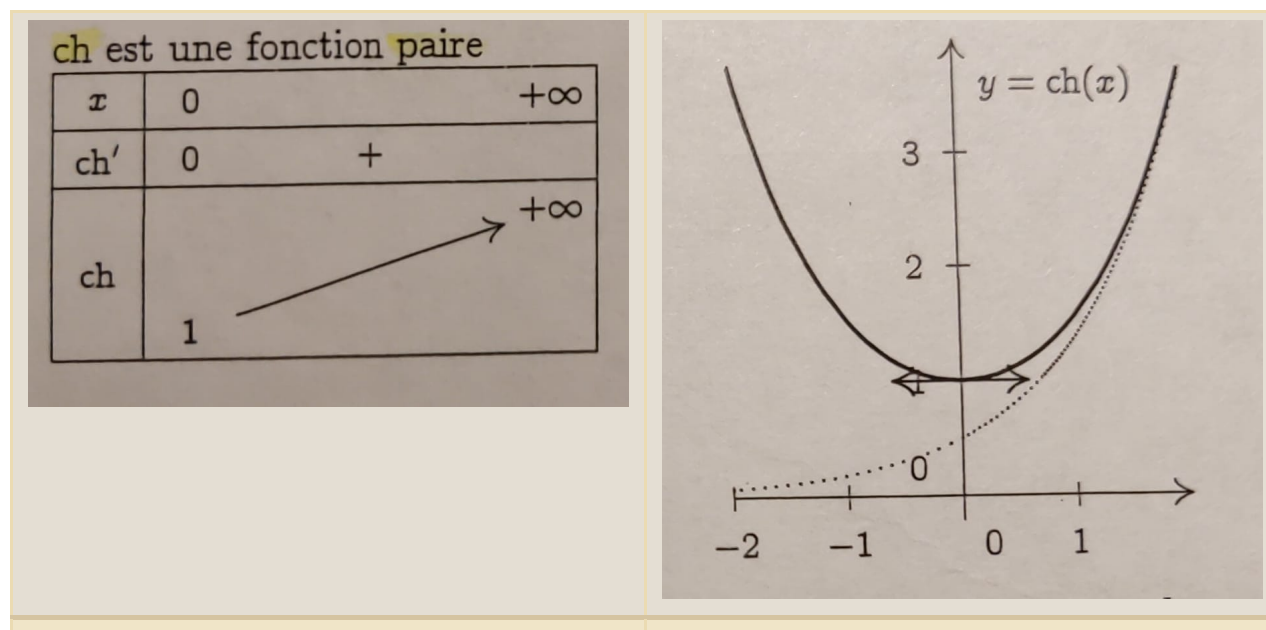
Cosinus et sinus hyperbolique

$$\forall x \in \mathbb{R}, \operatorname{ch}^2(x) - \operatorname{sh}^2(x) = 1$$

Cosinus hyperbolique

$$\operatorname{ch} : \mathbb{R} \longrightarrow \mathbb{R}; \quad \operatorname{ch}(x) = \frac{e^x + e^{-x}}{2}$$

- $\operatorname{ch}'(x) = \operatorname{sh}(x)$
- $\forall x \in \mathbb{R}, \operatorname{ch}(-x) = \operatorname{ch}(x)$



Sinus hyperbolique

$$\operatorname{sh} : \mathbb{R} \longrightarrow \mathbb{R}; \quad \operatorname{sh}(x) = \frac{e^x - e^{-x}}{2}$$

- $\operatorname{sh}'(x) = \operatorname{ch}(x)$
- $\forall x \in \mathbb{R}, \operatorname{sh}(-x) = -\operatorname{sh}(x)$

sh est une fonction impaire

x	0	$+\infty$
sh'	+	
sh	0	$+\infty$

